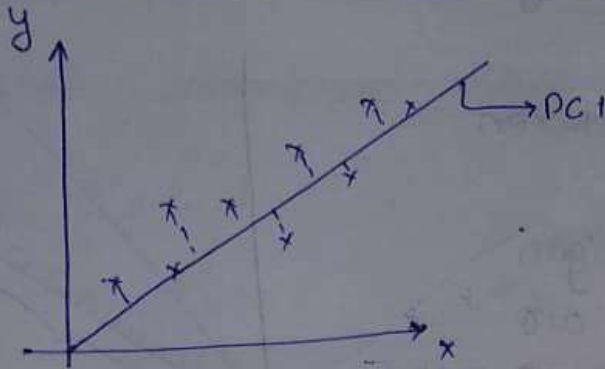
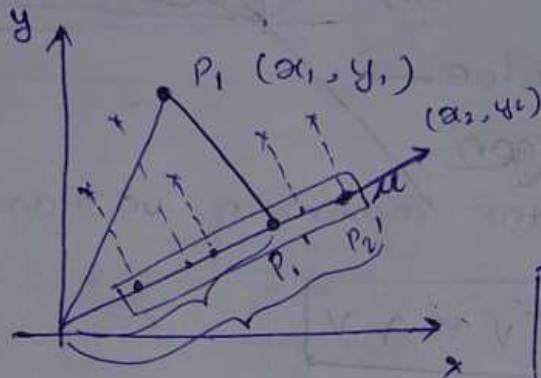


Math Intuition behind PCA Algorithm



2D $\rightarrow$ 1D

- ① Projection ✓
- ② Cost function $\rightarrow$ Variance ✓



$$\text{Proj}_{P_1} u = \frac{P_1 \cdot u}{\|u\|}$$

$\|u\| = 1 \rightarrow$ unit vector

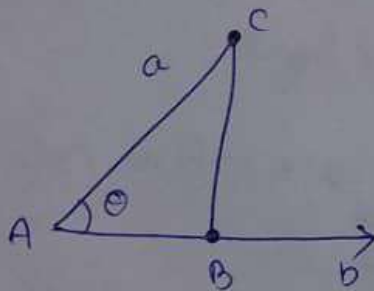
$$\text{Proj}_{P_1} u = P_1 \cdot u \Rightarrow \text{scalar value}$$

P_1'

$$P_0', P_1', P_2', P_3', P_4' \dots P_n'$$

scalar value

variance



$$P_0', P_1', P_2', P_3', P_4' \dots P_n'$$

$\Downarrow$

$$x_0', x_1', x_2', x_3', x_4' \dots x_n'$$

$$\text{max Variance} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$$

cost function

Goal:- Find the best unit vector which captures maximum variance

Eigen Vector And Eigen values

① Covariance Matrix between features

② Eigen vector and Eigen values will found out from this covariance matrix

③ Eigen vector $\rightarrow$ Eigen value $\rightarrow$ magnitude of the Eigen vector $\rightarrow$ capture the maximum variance

$$A v = \lambda v$$

Linear Transformation of matrix

